

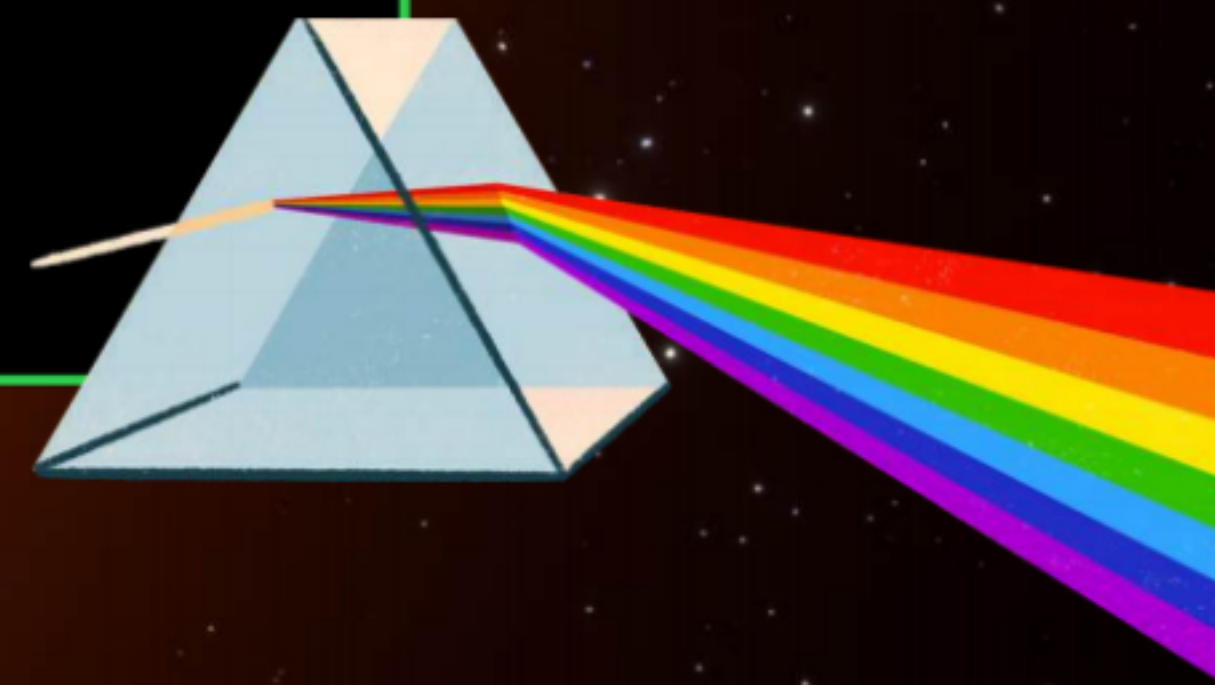
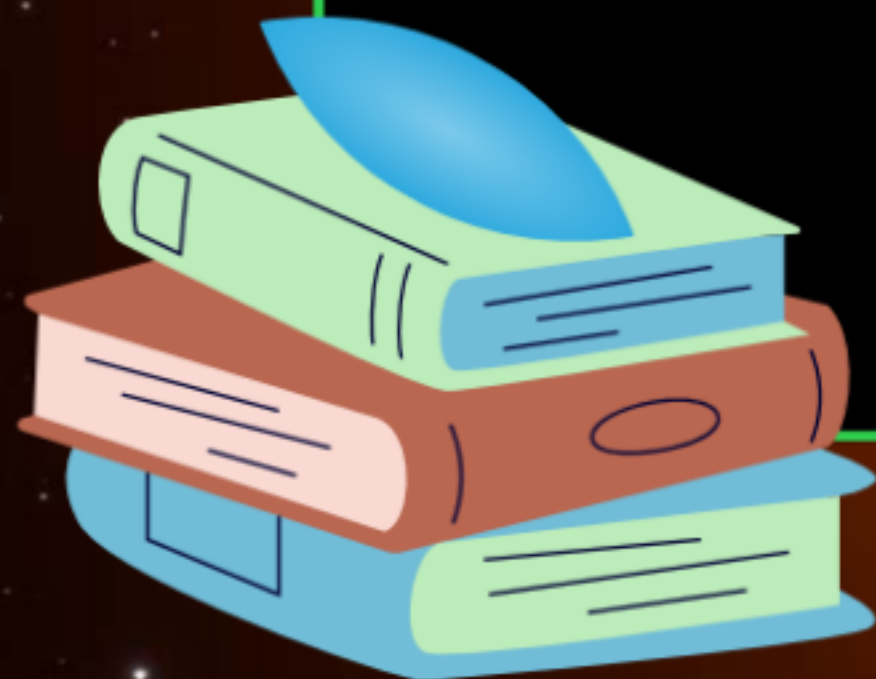
L3

आरंभ
BATCH

CLASS X - SCIENCE

LIGHT

PRASHANT KIRAD



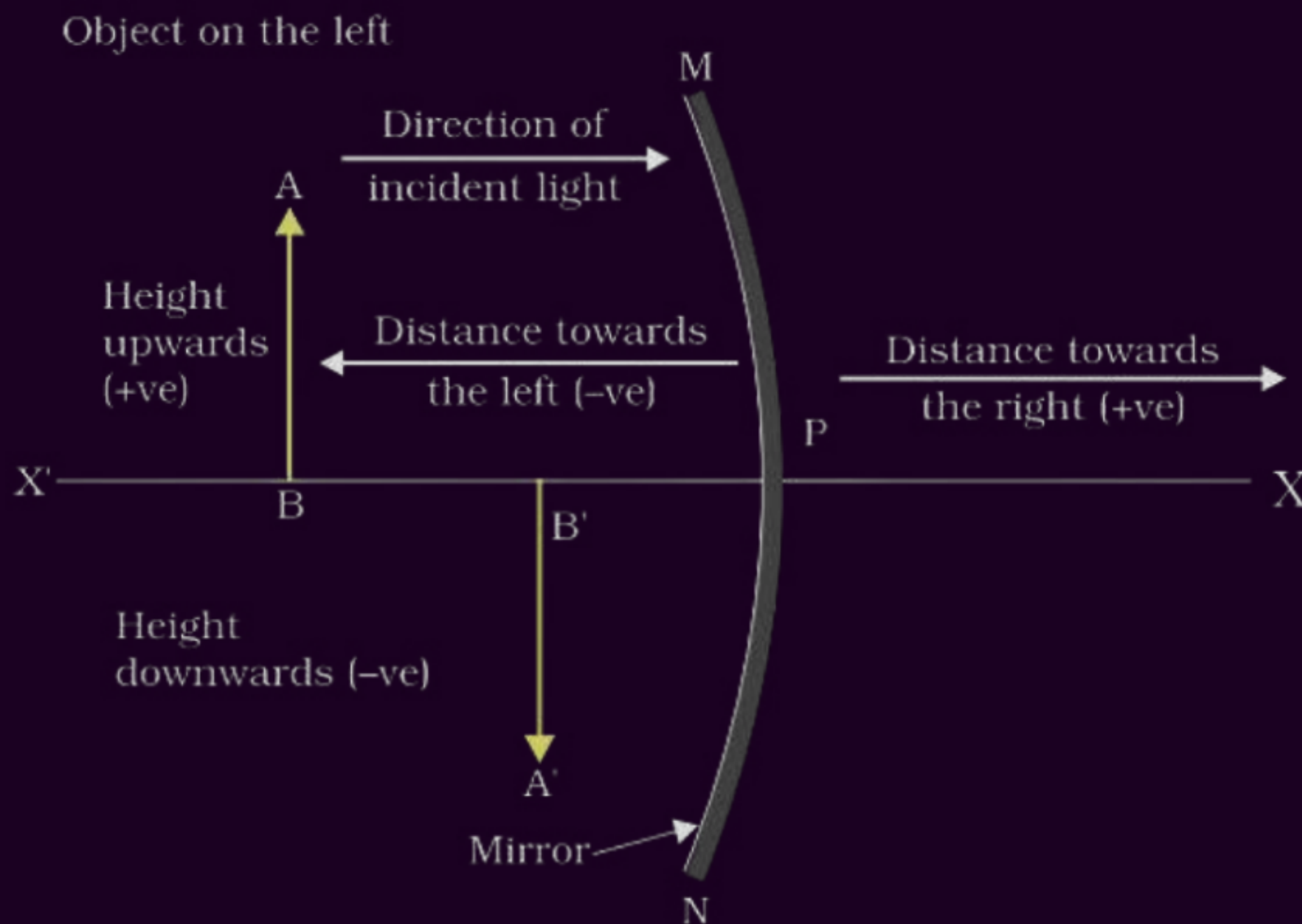
Sign Convention



Sign Convention in Mirrors

Basic Idea:

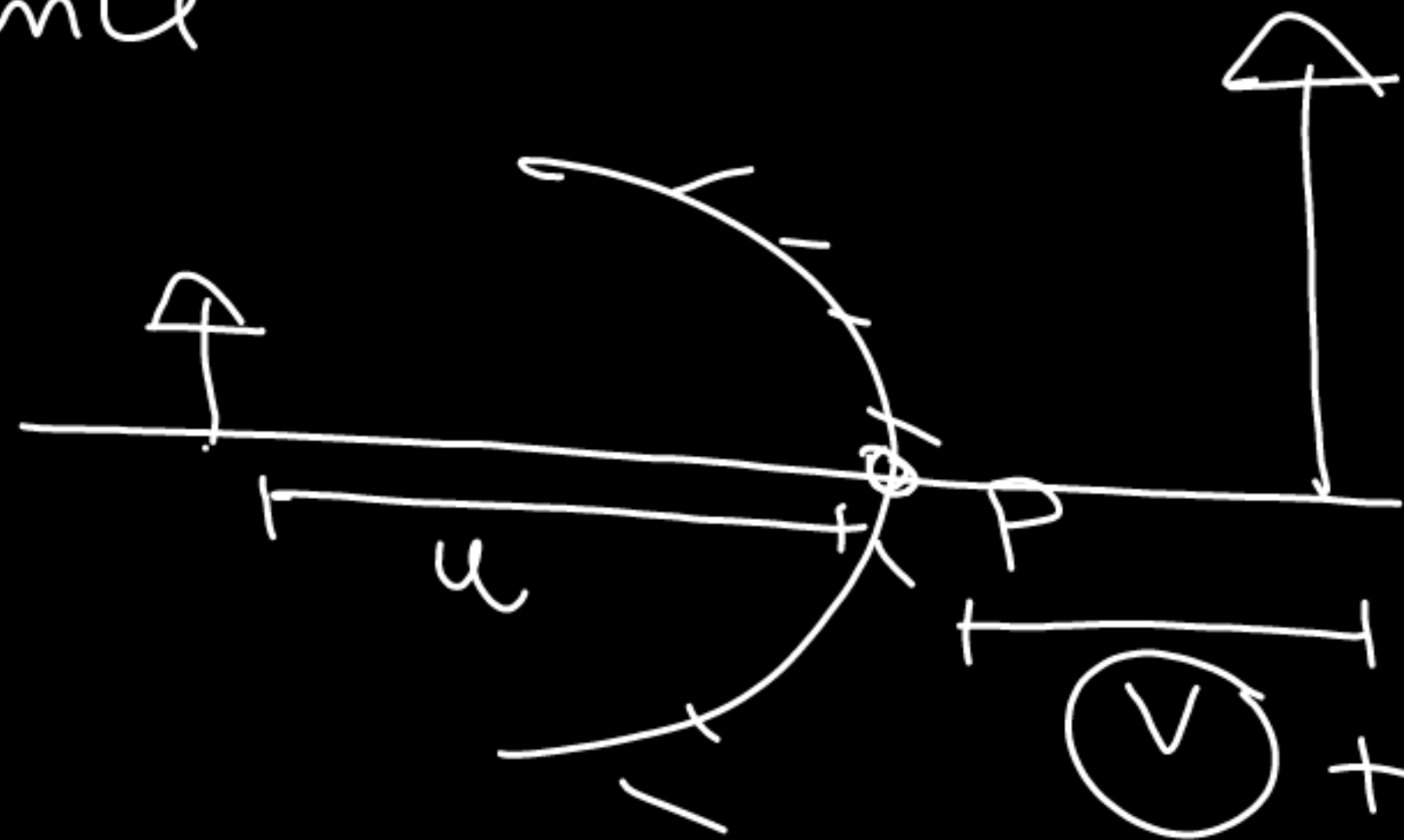
- The **pole (P)** of the mirror is taken as the origin.
- The **principal axis** is taken as the X-axis.
- Distances are measured from the **pole** of the mirror.

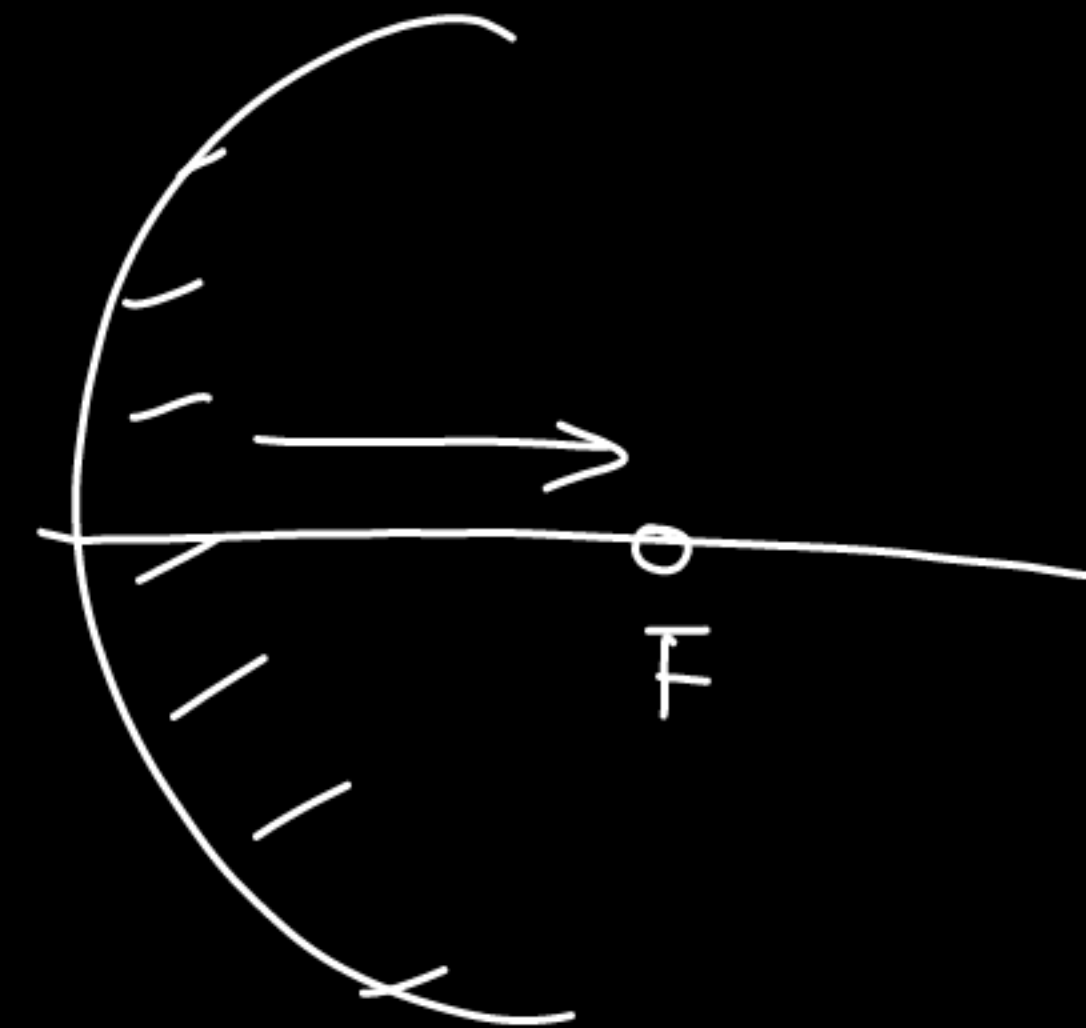
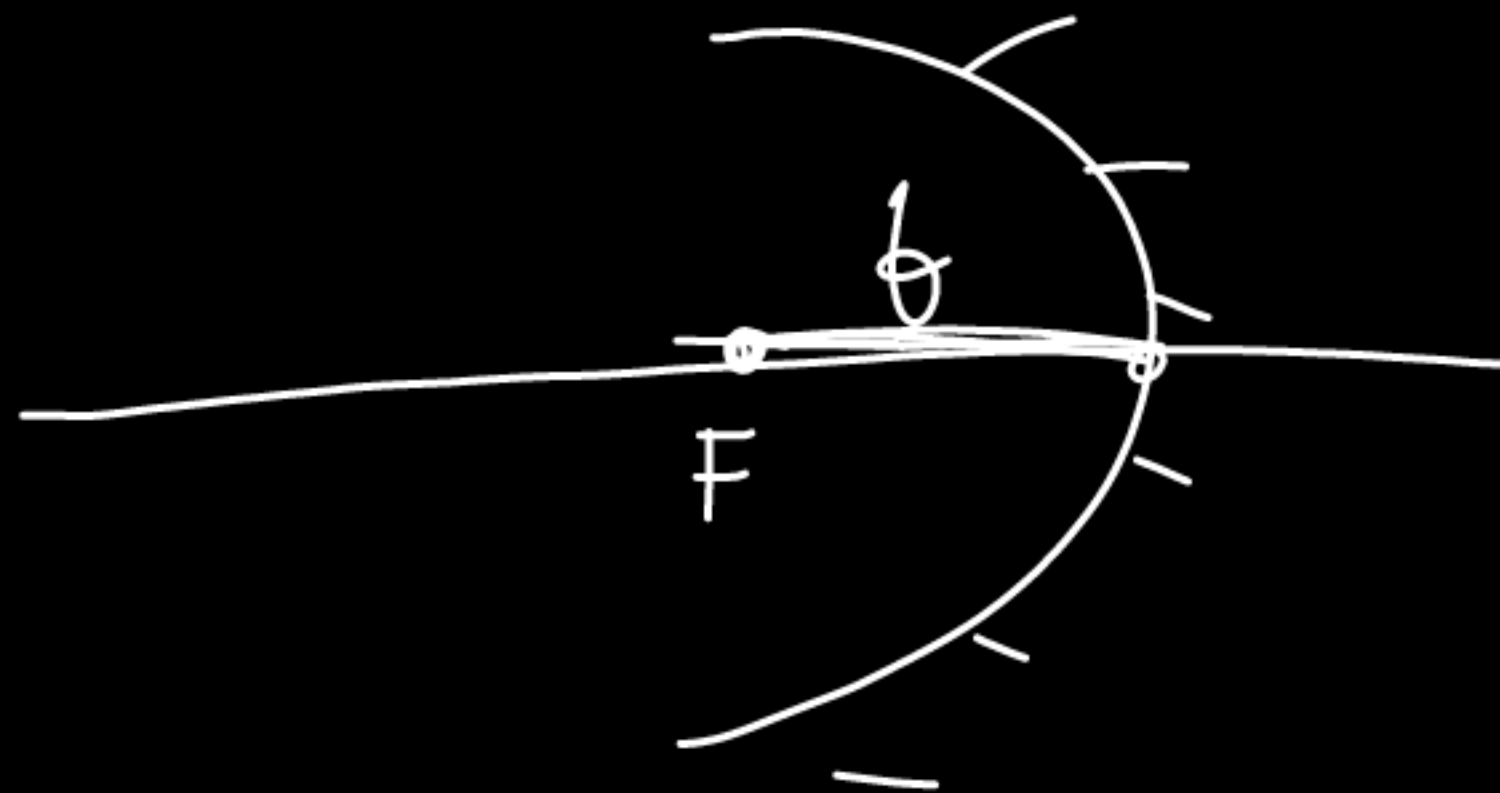


u → object distance

v → Image

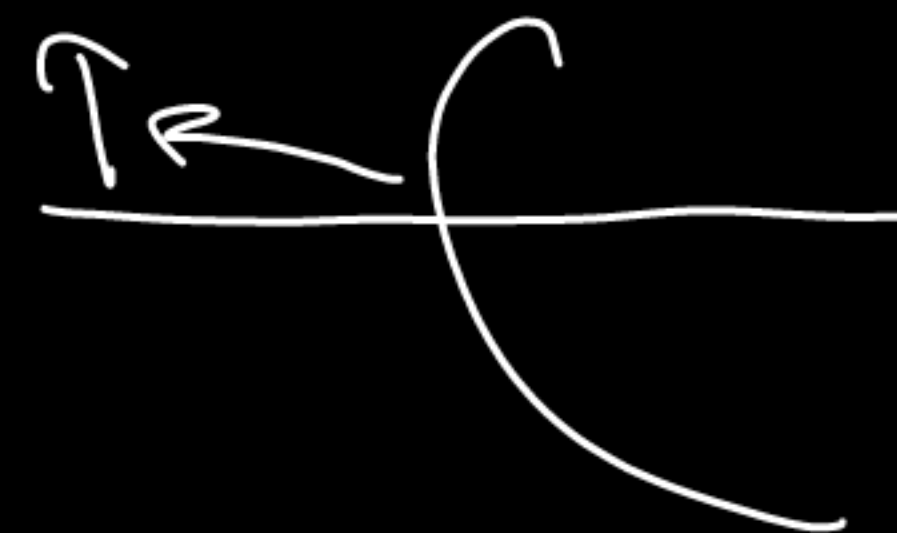
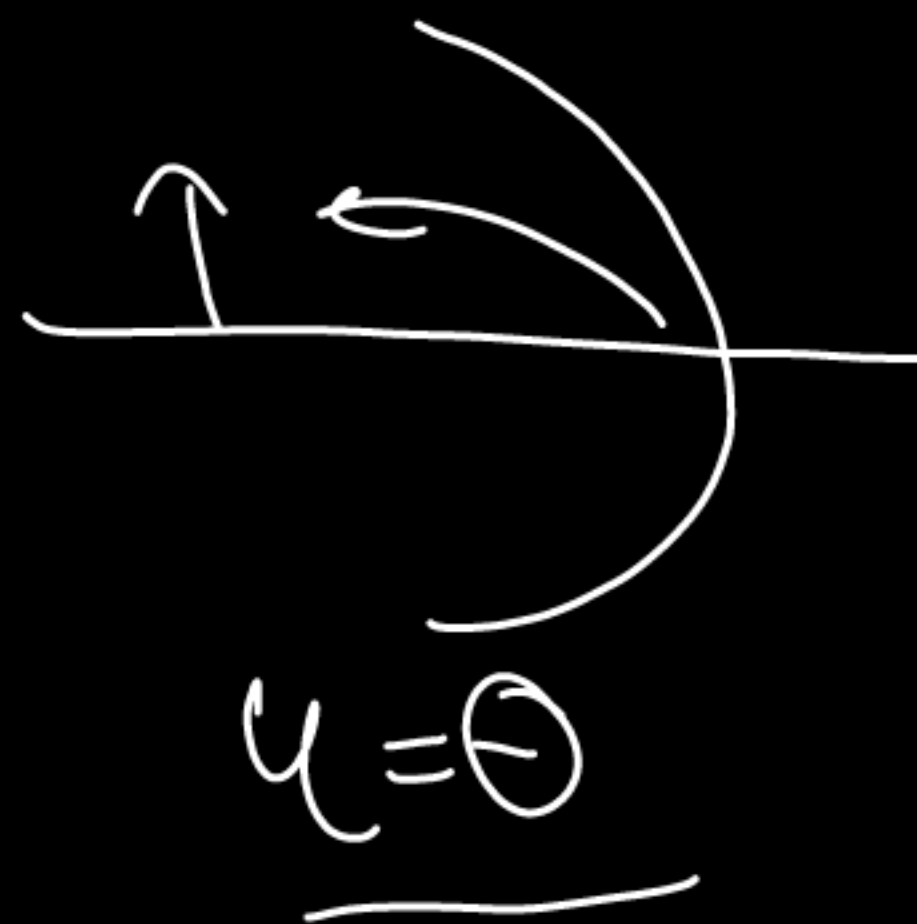
f → focal length

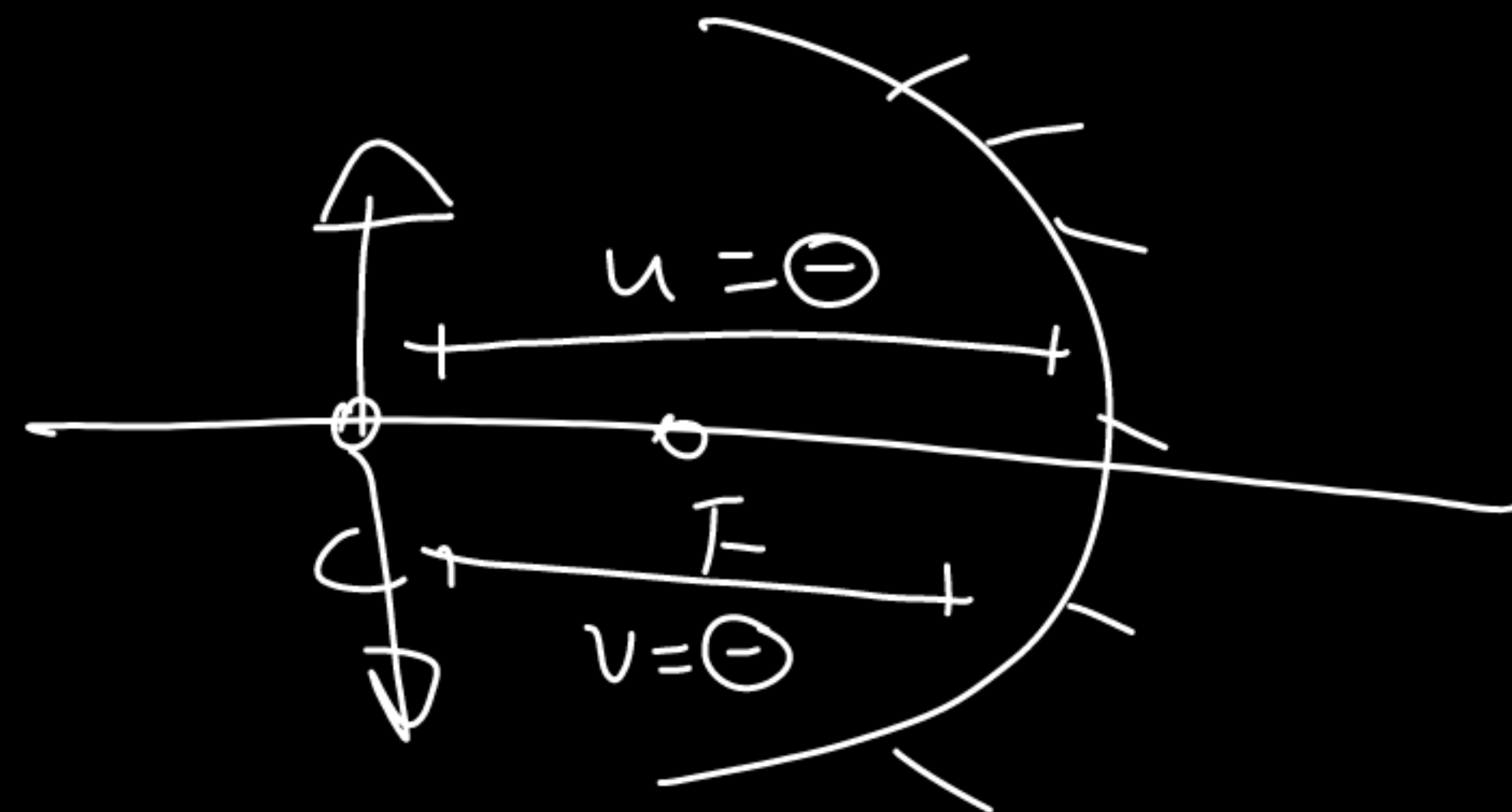


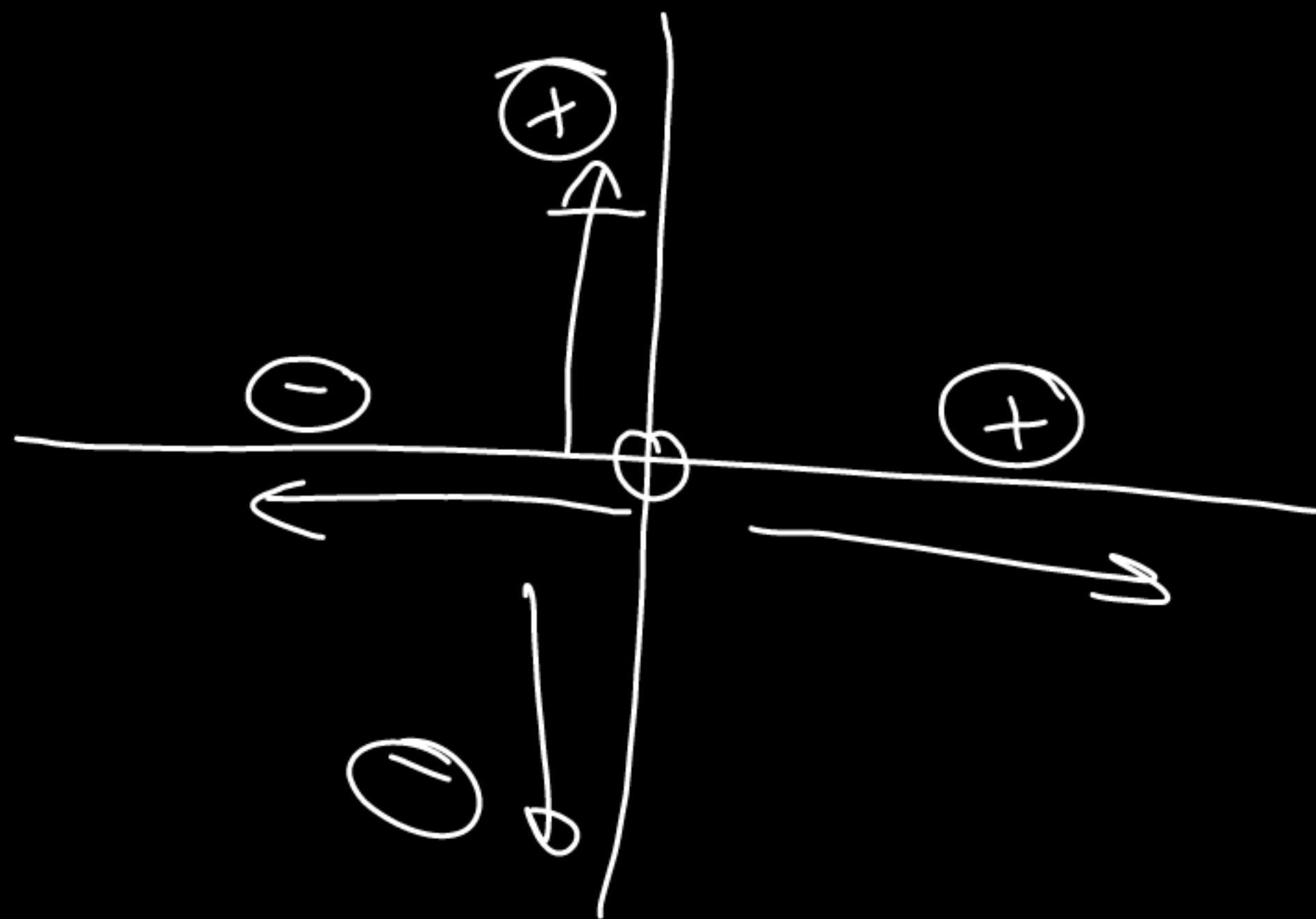


Concave $\rightarrow$ $\ominus$

$\oplus$ $\rightarrow$ convex





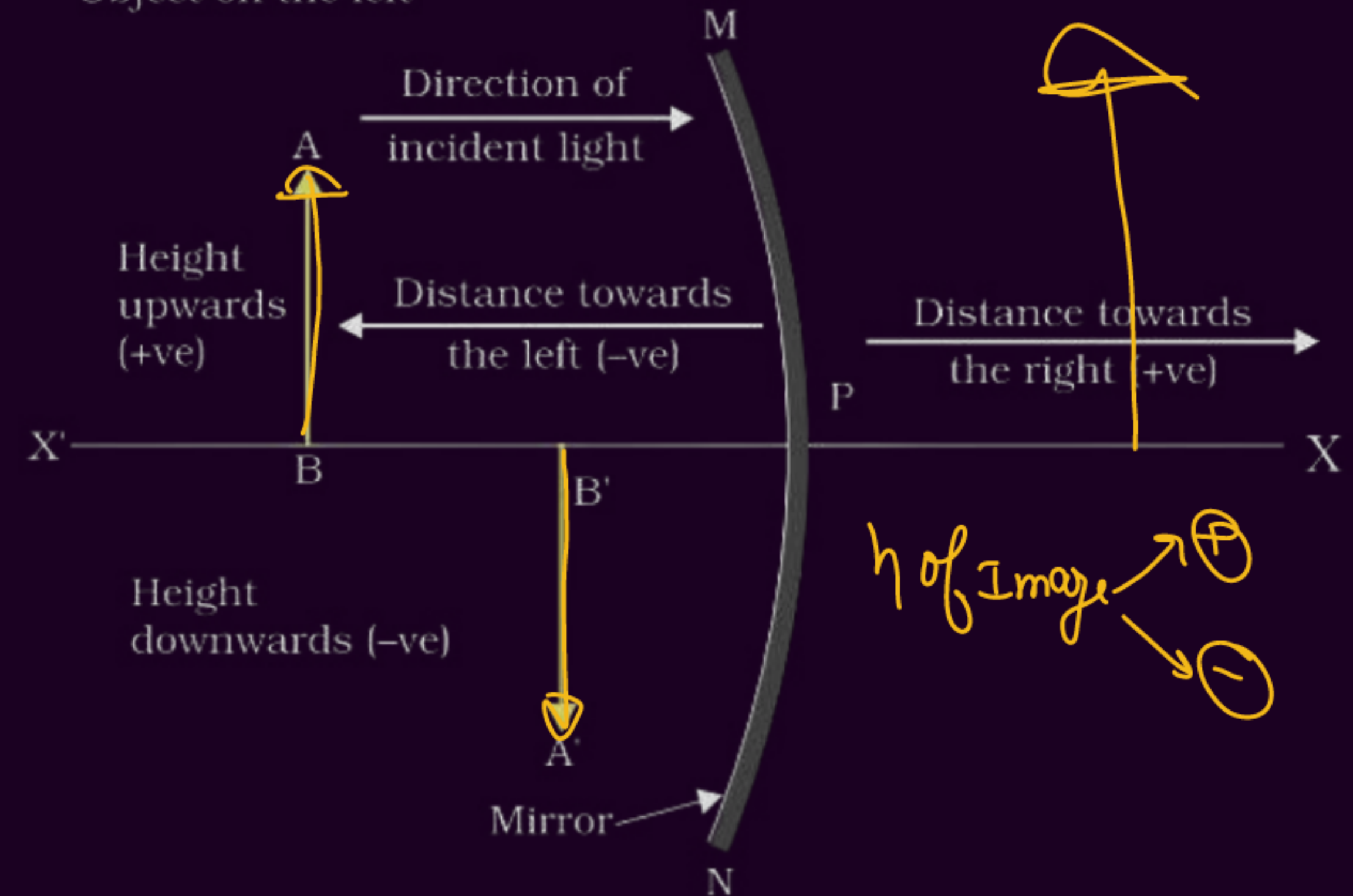


Sign Convention in Mirrors

Quantity / Direction	Sign
Distances measured from pole towards right side	Positive (+)
Distances measured from pole towards left side	Negative (-)
Heights measured above principal axis	Positive (+)
Heights measured below principal axis	Negative (-)
Object distance, u	Always negative (-)
Concave mirror focal length, f	Negative (-)
Convex mirror focal length, f	Positive (+)
Real image distance, v	Negative (-)
Virtual image distance, v	Positive (+)
Erect image height	Positive (+)
Inverted image height	Negative (-)

$h_{\text{of object}} = \oplus$

Object on the left



Important Formulae:

MIRROR FORMULAE

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

- The mirror formula establishes the connection between the object's distance (u), the image's distance (v), and the mirror's focal length (f).
- It is given by the equation:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

This formula applies to both plane mirrors and spherical mirrors, including convex and concave types.

Important Formulae:

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

MAGNIFICATION

- Magnification describes the ratio of the **height of the image (h')** to the **height of the object (h)**.
- It is represented by m and is calculated using:

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$m = \frac{\text{Height of the image } (h')}{\text{Height of the object } (h)}$$

$$m = \frac{h'}{h}$$

$$\text{Magnification } (m) = \frac{h'}{h} = -\frac{v}{u}$$



SOCHO!



Q. An object is placed at a distance of 30 cm from the pole of a concave mirror. If its real and inverted image is formed at 60 cm in front of the mirror, the focal length of the mirror is:

- (a) -15 cm
- ✓ (b) -20 cm
- (c) +20 cm
- (d) +15 cm

Given

$$\begin{aligned}
 u &= -30 \text{ cm} \\
 v &= -60 \text{ cm} \\
 f &= ?
 \end{aligned}$$

LCM

$$\begin{array}{r|l}
 2 & 30, 60 \\
 2 & 15, 30 \\
 15 & 15, 15
 \end{array}$$

$$\begin{aligned}
 \frac{1}{f} &= \frac{1}{u} + \frac{1}{v} \\
 &= -\frac{1}{30} + \left(-\frac{1}{60}\right)
 \end{aligned}$$

$$\frac{1}{f} = \frac{-2-1}{60} = \frac{-3}{60} = -\frac{1}{20}$$

$$f = -20 \text{ cm}$$



SOCHO!

Q. For a convex mirror the image distance (v) = 5 cm, focal length (f) = 10 cm and height of the image (h_i) = 7.5 cm. The correct representation according to sign conventions is:

- (a) $v = -5$ cm, $f = -10$ cm and $h_i = -7.5$ cm**
- (b) $v = -5$ cm, $f = +10$ cm and $h_i = -7.5$ cm**
- (c) $v = +5$ cm, $f = -10$ cm and $h_i = +7.5$ cm**
- (d) $v = +5$ cm, $f = +10$ cm and $h_i = +7.5$ cm**



SOCHO!

Q. To get an image of magnification -1 on a screen using a lens of focal length 20 cm , the object distance must be:

- (a) Less than 20 cm
- (b) 30 cm
- (c) 40 cm
- (d) 80 cm



SOCHO!

~~1~~ 2

Q. An optical device has been given to a student and he determines its focal length by focusing the image of the sun on a screen placed 24 cm from the device on the same side as the sun. Select the correct statement about the device.

- (a) Convex mirror of focal length 12 cm.**
- (b) Convex lens of focal length 24 cm.**
- (c) Concave mirror of focal length 24 cm.**
- (d) Convex lens of focal length 12 cm.**



SOCHO!

3

Q. Rays from the sun converge at a point 15 cm in front of a concave mirror. Where should an object be placed so that the size of its image is equal to the size of the object?

- (a) 15 cm in front of the mirror.**
- (b) 30 cm in front of the mirror.**
- (c) Between 15 cm and 30 cm in front of the mirror.**
- (d) More than 30 cm in front of the mirror.**



SOCHO!



Q. A 5 cm long rod is placed vertically in front of a concave mirror. If its 5 cm long image is formed at 20 cm in front of the mirror, the focal length of the mirror is:

- (a) -20 cm**
- (b) -10 cm**
- (c) -5 cm**
- (d) 20 cm**



SOCHO!



Q. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror.

2	15, 10
3	15, 5
5	5, 15

$$V = 6 \text{ cm}$$

$$u = -10 \text{ cm}$$

$$f = +15 \text{ cm}$$

$$v = ?$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{15} + \frac{1}{10} = \frac{2+3}{30} = \frac{5}{30} = \frac{1}{6} \quad \left[\frac{1}{15} - \left(-\frac{1}{10} \right) \right]$$



SOCHO!

Q. An object is placed at a distance of 10 cm in front of a concave mirror of focal length 15 cm. Use the mirror formula to determine the position of the image formed by this mirror. + Nature of Image

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= -\frac{1}{15} + \frac{1}{10} = \frac{-2+3}{30} = \frac{1}{30} \rightarrow +30\text{cm}$$

$$u = -10\text{cm}$$

$$f = -15\text{cm}$$

$$v = ?$$

$$m = -\frac{v}{u} = -\frac{30}{-10} = +3$$

$$m = ?$$



SOCHO!



Q. The magnification produced when an object is placed at a distance of 20 cm from a spherical mirror is $+1/2$. Where should the object be placed to reduce the magnification to $+1/3$?



SOCHO!



Q. If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use the mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer.



SOCHO!



Q. A concave mirror has a focal length of 20 cm. At what distance from the mirror should a 4 cm tall object be placed so that it forms an image at a distance of 30 cm from the mirror? Also calculate the size of the image formed.



SOCHO!

Q. An object 4 cm in height, is placed at 15 cm in front of a concave mirror of focal length 10 cm. At what distance from the mirror should a screen be placed to obtain a sharp image of the object. Calculate the height of the image

$$m = -\frac{v}{u} = -\left[\frac{+30}{+15}\right]$$

$$m = \boxed{-2}$$

Real → double

$$h_o = 4 \text{ cm}$$

$$u = -15 \text{ cm}$$

$$f = -10 \text{ cm}$$

$$v = ?$$

$$h_i = ?$$

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= -\frac{1}{10} + \frac{1}{15}$$

$$= \frac{-3+2}{30} = -\frac{1}{30}$$

$$\rightarrow v = -30 \text{ cm}$$

② very Important

$m = - \rightarrow$ negative $\rightarrow$ Real
 $+ \underline{I}$

$m = + \rightarrow$ Virtual $+ \in$

$m > 1 \rightarrow$ magnified

$m = 1 \rightarrow$ Same

$m < 1 \rightarrow$ Small

**Kal
Milte h!**